

Assessment of Efficient and Cost-Effective Vehicle Detection in Foggy Weather

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Abstract—Efficient and cost-effective vehicle detection under foggy and smoggy weather conditions remains a significant challenge in the domains of Intelligent Transportation Systems (ITS), Advanced Driver Assistance Systems (ADAS), Collision Avoidance Systems (CAS), and traffic surveillance. Vehicle detection in foggy weather, a critical requirement for preventing road collisions and ensuring driver assistance, has not been adequately evaluated. The state-of-the-art deep learning algorithm, YOLO-V11 network architecture, is studied and evaluated for efficient and cost-effective vehicle detection in diverse foggy weather road driving conditions. To our knowledge, YOLO-V11 performance has not been evaluated in diverse foggy weather road driving conditions. The vehicle Detection in Adverse Weather Nature (DAWN) and Foggy Driving (FD) datasets are utilized for the performance evaluation across diverse object detection performance metrics. The DAWN and FD datasets are customized and self-annotated to four vehicle detection classes: buses, cars, motorcycles, and trucks. Extensive experiments are performed to assess the efficiency and cost-effectiveness of YOLO-V11 models on DAWN and FD datasets. YOLO-V11s model outperformed with 73.1 % mAP and 71 % F1 score on the DAWN dataset and YOLO-V11m outperformed with 47.7 % mAP and 47 % F1 score. The recall and precision score of YOLO-V11m is higher on the DAWN dataset. Whereas, the recall score of the YOLO-V11l model is higher on the DAWN dataset and the precision score of YOLO-V11n is higher on the FD dataset. YOLO-V11 outperformed with an inference speed of 26 FPS on the DAWN dataset and YOLO-V11n outperformed with an inference speed of 13 FPS on the FD dataset. The comparative analysis concludes that the YOLO-V11n and YOLO-V11s models detect vehicles with faster inference speeds while maintaining high accuracy. Conversely, YOLO-V11m and YOLO-V11l models detect vehicles with enhanced accuracy but at the cost of slower inference speeds.

Index Terms—Vehicle Detection, DAWN, FD, mAP50, FPS, Foggy Weather.

I. INTRODUCTION

Computer vision aims to equip computers to recognize and understand visual elements, similar to human vision. Computer vision-based object detection algorithms use the computational power of modern high-end Central Processing Unit (CPU) and Graphical Processing Unit (GPU) systems, which are used for traffic surveillance, autonomous vehicles, augmented reality, and manufacturing [1]. Object detection algorithms use

camera sensors to locate and classify objects in indoor and outdoor environments. Because the cameras are now more accurate at detecting objects and are more cost-effective than other sensors [2]. A Convolutional Neural Network (CNN) comprises neurons with learnable weights and biases. The network includes an input, output, and multiple hidden layers. The hidden layers typically consist of convolutional, pooling, fully-connected, and various normalization layers [3]. You Only Look Once (YOLO) transforms object detection into a regression problem. YOLO uses a single CNN to generate object bounding boxes and directly predict class-level probabilities from the input images. Primarily, the YOLO algorithm divides the input images into grid cells, predicting class-level probabilities, confidence scores, and bounding boxes, and finally, detecting objects based on IOU thresholds [4] as shown in Fig. 1.

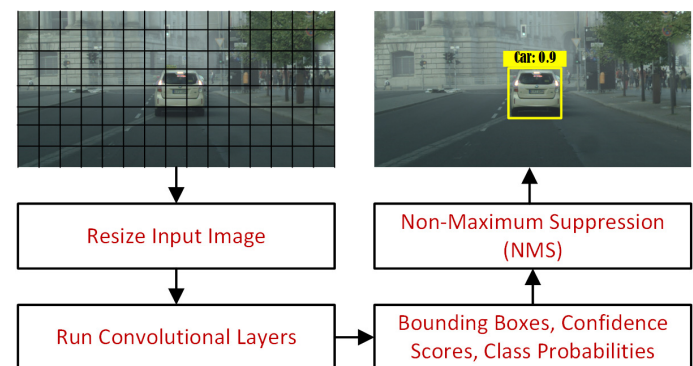


Fig. 1. YOLO Working

The main research objectives and contributions of the proposed research work are:

- To explore the architectural components of the state-of-the-art YOLO-V11 algorithm.
- To propose a research methodology to assess the efficiency and cost-effectiveness of YOLO-V11 algorithmic models: YOLO-V11n, YOLO-V11s, YOLO-V11m, and YOLO-V11l.
- To train, validate, and test the YOLO-V11 algorithmic models using customized and self-annotated DAWN and FD foggy weather vehicle detection datasets.

- To compare and analyze the vehicle detection results.

The overall organization of the paper is as follows. Section I provides a concise introduction to vehicle detection in foggy weather. Section II presents a review of the literature. Section III, presents the research methodology, dataset selection and pre-processing, training and validation parameters setup, and performance metrics. Section IV provides the training and validation results. The comparison and analysis of results are also discussed. Section V concludes the research findings and highlights the future work directions.

II. LITERATURE REVIEW

Vehicle detection in normal weather is evaluated extensively using single-stage and two-stage deep learning algorithms on diverse datasets. A research gap still exists for efficient and cost-effective vehicle detection in foggy weather conditions. The YOLO series of algorithms has emerged as a potential solution for detecting vehicles at higher inference speeds with reasonable accuracy. The YOLO series of algorithms has evolved into many versions. The YOLO version 11 (YOLO-V11) architecture is mainly based on YOLO version 5 (YOLO-V5) and YOLO version 8 (YOLO-V8) algorithms.

The YOLO-V11 network architecture comprises a backbone network, neck, and head. The backbone of YOLO-V11 consists of convolutional layers, C3k2 (faster implementation of Cross Stage Partial (CSP) bottlenecks with 2 convolutional layers) layers, Spatial Pyramid Pooling Fast (SPPF) module, and C2PSA (2 convolutional layers with Position Sensitive attention (PSA) mechanism) module. The backbone of YOLO-V11 efficiently extracts the spatial features for enhanced feature extraction, processing, and pooling of features at diverse scales. The neck aggregates the features using bottom-up and top-down paths towards the detection head. The detection head detects the objects using pre-defined anchor boxes. The main difference between YOLO-V5, YOLO-V8, and YOLO-V11 is bottleneck modules. YOLO-V5 contains C3 bottleneck modules, YOLO-V8 contains C2f modules, and YOLO-V11 contain C3k2 modules. The architectural diagram of the YOLO-V11 algorithm is illustrated in Fig. 2.

A comprehensive overview of the related work is discussed in this section. A diverse set of experiments was performed to evaluate the vehicle detection efficiency and cost-effectiveness of YOLO-V5, YOLO-V5 variants, and YOLO-V8 algorithms on DAWN and FD datasets. The proposed YOLO-V5 variants were based on Common Block Attention Module (CBAM), Normalized Attention Module (NAM), and Simplified Attention Module (SimAM) modules integrated into the backbone of the baseline YOLO-V5 algorithm. They also evaluated the performance of YOLO-V5l with the Feature Pyramid Pooling (FPN) path [5].

Another study evaluated the performance of the YOLO-V5 algorithm in adverse weather conditions [6]. The YOLO-V5n model is on the DAWN dataset using Google Colab with fog, transfer learning, fog with down-sampling, and fog with down-sampling with transfer learning. The mAP50 score on DAWN-fog is 88.3 %, DAWN-fog with transfer learning is 84.4 %, and DAWN-fog with down-sampling is 85 %, and DAWN-fog with down-sampling and transfer learning is 78.3 % [7].

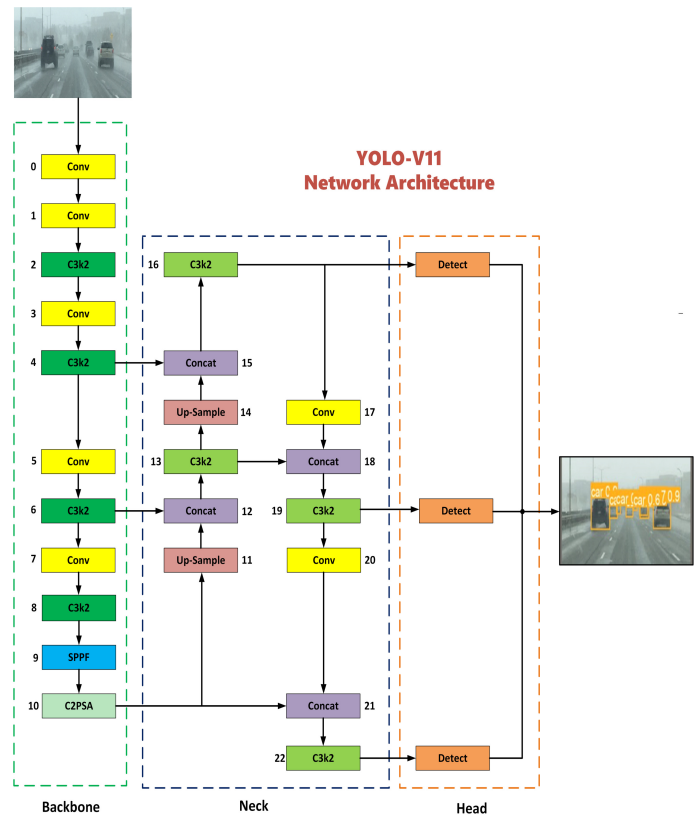


Fig. 2. YOLO-V11 network architecture

DAWN-fog with down-sampling is 85 %, and DAWN-fog with down-sampling and transfer learning is 78.3 % [7].

III. RESEARCH METHODOLOGY

The steps of the research methodology consist of dataset selection and preprocessing, vehicle detection algorithm selection, experimental parameters, and performance metrics. The research methodology is illustrated in Fig. 3. The Google Colab platform is utilized for Google Drive integration, YOLO-V11 code and datasets management, a rich set of resource utilization, and results storage. There are four training and four validation experiments using YOLO-V11 algorithm models performed separately for both DAWN and FD datasets. The details of dataset selection and pre-processing, experimental parameters, and performance metrics are discussed in the subsequent sections.

A. Datasets Selection and Pre-Processing

The state-of-the-art DAWN [8] [9] and FD [10] [11] datasets are selected for vehicle detection in foggy weather. The DAWN dataset consists of 1000 foggy images and the FD dataset consists of 101 foggy images in a diverse fog density for normal to highway driving scenarios. These datasets are self-annotated and customized for vehicle detection for buses, cars, motorcycles, and trucks. The holdout method is utilized for the training and validation images split with a ratio of 80 % training and 20 % validation images. The state-of-the-art

YOLO-V11 single-stage deep learning algorithm along with nano, small, medium, and large models are selected for the vehicle detection task.

The detailed list of experimental parameters is summarized in Table I. The diverse performance metrics are selected to evaluate the vehicle detection performance of the YOLO-V11 [12] algorithm and its models.

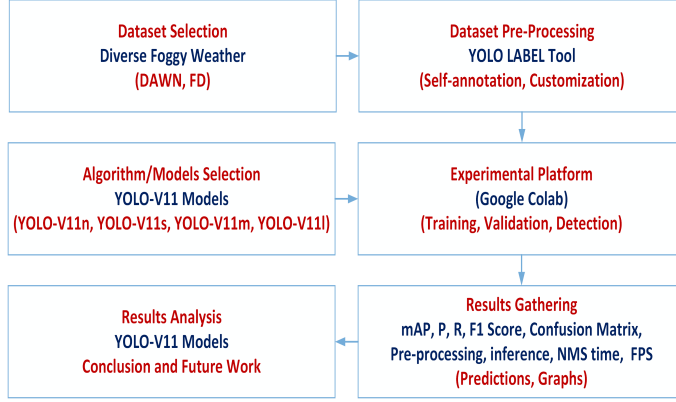


Fig. 3. Research Methodology

B. Performance metrics

The state-of-the-art performance evaluation metrics are utilized to assess the efficiency and cost-effectiveness of the YOLO-V11 algorithm and its models [13]. A description of these metrics is provided below.

1) *Confusion matrix*: The confusion matrix shows the number of True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values of buses, cars, trucks, and motorcycles. These values help to drive the relationship between Ground Truth (GT) and the Predicted Bounding Box (PBB) values.

2) *Precision (P)*: The vehicle detection model efficiency of correctly identifying the significant vehicle objects only is known as model precision. A higher value of precision is preferable. The mathematical equation of precision is expressed below.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

3) *Recall (R)*: The vehicle detection models effectiveness of identifying all vehicle objects. A higher value of recall is preferable. The mathematical equation of recall is expressed below.

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

4) *Average Precision*: The vehicle detection model performance is measured by taking the average of precision scores over recall scores under the precision-recall (PR) curve. The higher AP results in good performance. The mathematical equation of AP is expressed below.

$$AP = \frac{1}{11} \sum_{re=0,0.1,0.2,\dots,1} P_{interpolation}(r) \quad (3)$$

5) *Mean Average Precision (mAP)*: The vehicle detection model's accuracy of correctly identifying all vehicle classes in a specific dataset. The higher mAP scores result in higher accuracy. The mathematical equation of mAP is expressed below.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4)$$

6) *F1 score*: The vehicle detection models the harmonic mean of precision and recall. The mathematical equation of the F1 score is expressed below.

$$F1score = 2 \times \frac{P * R}{P + R} \quad (5)$$

7) *Frame Per Second (FPS)*: The inference speed of vehicle detection is measured in FPS. The pre-processing time, inference time, and post-processing time influence the FPS rate of the vehicle detection algorithm. The higher the value of FPS, the higher the vehicle detection speed will be.

IV. RESULTS AND DISCUSSION

The efficient and cost-effective vehicle detection performance regarding mAP50 and FPS rate in foggy weather is evaluated and compared using the YOLO-V11 algorithm on DAWN and FD datasets.

A. Comparison of efficient vehicle detection results on DAWN and FD datasets

The YOLO-V11 algorithm vehicle detection performance is evaluated and compared on DAWN and FD datasets in foggy weather. The YOLO-V11n model detects cars with a mAP50 score of 85.6 % on the DAWN dataset and 69.7 % on the FD dataset. The mAP50 score of bus detection is 31.9 % on the DAWN dataset and 31.1 % on the FD dataset. The mAP50 score of truck detection is 69.7 % on the DAWN dataset and 26.2 % on the FD dataset. The mAP50 score of motorcycle detection is 51.8 % on the DAWN dataset and no motorcycle object is detected on the FD dataset. The multi-class mAP50 score of vehicle detection is 59.7 % on the DAWN dataset and 42.3 % on the FD dataset.

The multi-class mAP50-95, F1 score, precision, and recall results are summarized in Table II and Table IV respectively. In the second set of experiments, YOLO-V11s model performance is evaluated. The mAP50 score of car detection is 88.2 % on the DAWN dataset and 67.4 % on the FD dataset. The mAP50 score of bus detection is 36.3 % on the DAWN dataset and 41.9 % on the FD dataset. The mAP50 score of truck detection is 68.4 % on the DAWN dataset and 17 % on the FD dataset. The mAP50 score of motorcycle detection is 99.5 % on the DAWN dataset and no motorcycle object is detected on the FD dataset. The multi-class mAP50 score of vehicle detection is 73.1 % on the DAWN dataset and 42.1 % on the FD dataset.

The multi-class mAP50-95, F1 score, precision, and recall results are summarized in Table II and Table IV respectively. The visualization of mAP, P, R, and loss functions results of

TABLE I
EXPERIMENTAL PARAMETERS (TRAINING AND VALIDATION)

Parameter(s)	Value(s)
YOLO-V11 Models	YOLO-V11n, YOLO-V11s, YOLO-V11m, YOLO-V11l
Datasets	DAWN, FD
Classes	4 (Buses, Cars, Motorcycles, Trucks)
Python and PyTorch Version	Python-3.10.12, Torch-2.5.0+cu121 with CUDA enabled
Ultralytics Version	8.3.27
Hardware and GPU Specifications	2 CPUs, Tesla T4 GPU (15GB), Systems RAM 12.7GB, Disk Space 112.6 GB
Experimental Platform	Google Colab
Image Size	640 train, 640 validation
Batch Size (Traning)	16
Batch Size (Validation)	1
Training Epochs	300
Data Loaders/Workers	2
Hyperparameters	Default
Optimizer	AdamW(lr=0.00125, momentum=0.9)

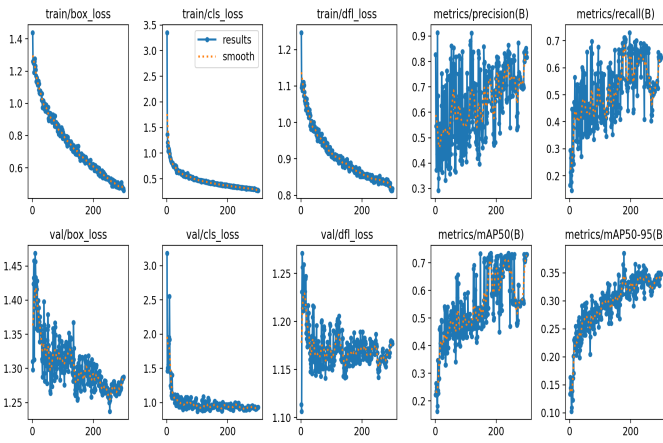


Fig. 4. YOLO-V11s graphical results on DAWN dataset

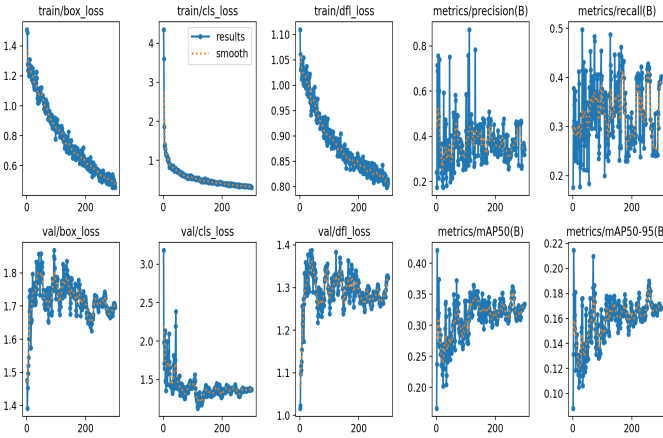


Fig. 5. YOLO-V11s graphical results on FD dataset

YOLO-V11s on DAWN and FD datasets is shown in Fig. 4 and Fig. 5 respectively. The vehicle detection and predictions of YOLO-V11s on the DAWN and FD datasets are shown in Fig. 6 and Fig. 7 respectively.

In the third set of experiments, YOLO-V11m performance is evaluated. The mAP50 score of car detection is 89.1 % on the DAWN dataset and 68.7 % on the FD dataset. The mAP50 score of bus detection is 33.2 % on the DAWN dataset and 35.6 % on the FD dataset. The mAP50 score of truck detection is 69.8 % on the DAWN dataset and 38.6 % on the FD dataset. The mAP50 score of motorcycle detection is 99.5 % on the DAWN dataset and no motorcycle object is detected on the FD dataset. The multi-class mAP50 score of vehicle detection is 72.9 % on the DAWN dataset and 47.7 % on the FD dataset.

The multi-class mAP50-95, F1 score, precision, and recall results are summarized in Table II and Table IV respectively. Finally, YOLO-V11l performance is evaluated. The mAP50 score of car detection is 87.9 % on the DAWN dataset and 66.4 % on the FD dataset. The mAP50 score of bus detection is 36.3 % on the DAWN dataset and 41.2 % on the FD dataset. The mAP50 score of truck detection is 67.2 % on the DAWN dataset and 29.3 % on the FD dataset. The mAP50 score of motorcycle detection is 99.5 % on the DAWN dataset and no motorcycle object is detected on the FD dataset. The multi-class mAP50 score of vehicle detection is 72.7 % on the DAWN dataset and 45.6 % on the FD dataset.

The multi-class mAP50-95, F1 score, precision, and recall results are summarized in Table II and Table IV respectively.

B. Comparison of cost-effective vehicle detection results on DAWN and FD datasets

The YOLO-V11 algorithm vehicle detection inference speed (FPS rate) is validated with batch size 1 on DAWN and FD datasets in terms of FPS rate in foggy weather.

The pre-processing time of YOLO-V11n is less on both DAWN and FD datasets as compared to other YOLO-V11 models. The inference time of YOLO-V11s is less on both DAWN and FD datasets as compared to other YOLO-V11 models. The post-processing time of YOLO-V11m is small on both DAWN and FD datasets as compared to other YOLO-V11 models. The inference speed of YOLO-V11n is 21 FPS on the

TABLE II
VEHICLE DETECTION RESULTS (ACCURACY) OF YOLO-V11 ON DAWN DATASET

Model	% mAP50 Car	% mAP50 Bus	% mAP50 Truck	% mAP50	% mAP50-95	% F1 score	% P	% R
YOLO-V11n	85.6	31.9	69.7	59.7	36.1	55	63.7	59.3
YOLO-V11s	88.2	36.3	68.4	73.1	38.5	71	73.8	69.1
YOLO-V11m	89.1	33.2	69.8	72.9	38.3	71	75.7	69.5
YOLO-V11l	87.9	36.3	67.2	72.7	37.2	69	80	63.7

TABLE III
VEHICLE DETECTION RESULTS (SPEED) OF YOLO-V11 ON DAWN DATASET

Model	Pre-processing Time (ms)	Inference Time (ms)	Post-processing Time (ms)	FPS (Inference Speed)
YOLO-V11n	0.4	30.4	18	21
YOLO-V11s	0.6	26.5	11.1	26
YOLO-V11m	0.3	29.5	9.4	26
YOLO-V11l	0.9	42.4	17.7	16

TABLE IV
VEHICLE DETECTION RESULTS (ACCURACY) OF YOLO-V11 ON FD DATASET

Model	% mAP50 Car	% mAP50 Bus	% mAP50 Truck	% mAP50	% mAP50-95	% F1 score	% P	% R
YOLO-V11n	69.7	31.1	26.2	42.3	22.5	35	80.9	29.3
YOLO-V11s	67.4	41.9	17	42.1	21.5	37	71.1	37.8
YOLO-V11m	68.7	35.6	38.6	47.7	26.9	47	60.7	38.6
YOLO-V11l	66.4	41.2	29.3	45.6	24.7	45	70.9	37.1

TABLE V
VEHICLE DETECTION RESULTS (SPEED) OF YOLO-V11 ON FD DATASET

Model	Pre-processing Time (ms)	Inference Time (ms)	Post-processing Time (ms)	FPS (Inference Speed)
YOLO-V11n	1.8	34.6	41.4	13
YOLO-V11s	1.8	30.1	55.1	12
YOLO-V11m	2.8	39.8	38.4	12
YOLO-V11l	2.0	47	38.6	11

DAWN dataset and 13 FPS on the FD dataset. The inference speed of YOLO-V11s is 26 FPS on the DAWN dataset and 12 FPS on the FD dataset. The inference speed of YOLO-V11m is 26 FPS on the DAWN dataset and 12 FPS on the FD dataset. Lastly, the inference speed of YOLO-V11l is 16 FPS on the DAWN dataset and 11 FPS on the FD dataset. The pre-processing time, inference time, and NMS time along with the FPS rate of the YOLO-V11 algorithm on DAWN and FD datasets are summarized in Table III and Table V respectively.

V. CONCLUSION AND FUTURE WORK

Vehicle detection in diverse foggy weather road driving conditions is a challenging problem for autonomous vehicles, driver assistance, collision avoidance, and traffic surveillance systems.

The state-of-the-art YOLO-V11 algorithm robustness is evaluated in diverse foggy weather datasets. Numerous experiments are performed on YOLO-V11n, YOLO-V11s, YOLO-V11m, and YOLO-V11l models to assess the efficiency in terms of precision, recall, mAP, and F1 scores; and cost-effectiveness in terms of pre-processing, inference, NMS time as well as FPS rate. YOLO-V11s model outperformed with 73.1 % mAP

and 71 % F1 score on the DAWN dataset and YOLO-V11m outperformed with 47.7 % mAP and 47 % F1 score. The recall and precision score of YOLO-V11m is higher on the DAWN dataset. Whereas, the recall score of the YOLO-V11l model is higher on the DAWN dataset and the precision score of YOLO-V11n is higher on the FD dataset. YOLO-V11 outperformed with an inference speed of 26 FPS on the DAWN dataset and YOLO-V11n outperformed with an inference speed of 13 FPS on the FD dataset. It is concluded that YOLO-V11n and YOLO-V11s models are less efficient but more cost-effective. Similarly, the YOLO-V11m and YOLO-V11l models are more efficient but less cost-effective.

The efficiency and cost-effectiveness of the YOLO-V11 algorithm on more diverse datasets of foggy cityscapes, UA-DETRAC, and RTTS in more diverse foggy weather road driving conditions will assessed in future work. Furthermore, the customizations and modifications in the architecture of YOLO-V11 have enormous potential to achieve more accuracy with a higher inference speed. The proposed solutions may incorporate attention mechanisms, diverse loss functions, vision transformers, and ghost convolutions among other techniques.



Fig. 6. YOLO-V11s predictions on DAWN dataset



Fig. 7. YOLO-V11s predictions on FD dataset

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